

# SnapTarget

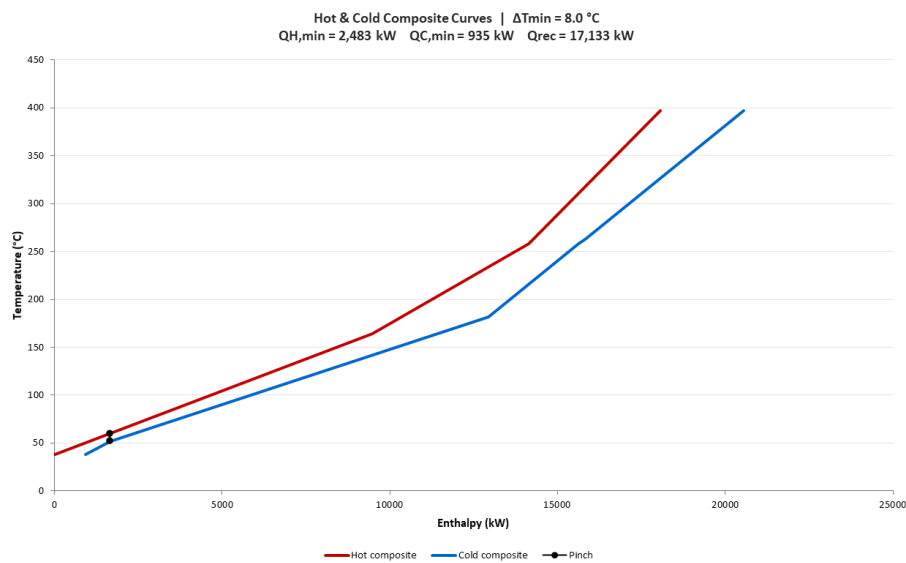
HEN Supertargeting Add-in for Microsoft Excel — User Guide

## User Guide

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Heat Exchanger Network synthesis for a Kerosene Hydrotreater — Graduation Project



SnapTarget performs pinch-analysis supertargeting inside Excel: energy, area and units targets (Problem Table Algorithm + Bath formula), the  $\Delta T_{min}$  trade-off study with two capital-cost models, and seven live charts (one per sheet). It is a pure-VBA port of the validated TOP v9 model and reproduces it to machine precision (worst relative deviation  $4 \times 10^{-15}$  over the full sweep).

## 1. Installation

Try it once: double-click SnapTarget .xlam and enable macros when prompted. Install permanently: Excel → File → Options → Add-ins → Manage: Excel Add-ins → Go... → Browse... → select the file. If Windows blocks it, right-click the file → Properties → Unblock.

Workbooks created with SnapTarget use its worksheet functions, so they display #NAME? on machines where the add-in is not installed.

## 2. Workflow

| Step | Ribbon button   | What you get                                                                                                                                                       |
|------|-----------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1    | New Template    | Streams input workbook, pre-filled with the kerosene hydrotreater demo case (streams CMIX, C3, C4, H1–H3 + economics), plus Credits and License (EULA) sheets.     |
| 2    | —               | Set $\Delta T_{min}$ (B4) and No. of streams (E4) — the stream table instantly resizes to that many rows — then enter the streams and adjust the four input boxes. |
| 3    | Compute Targets | PTA sheet (full heat cascade, pinch highlighted) and the live Results sheet — three sections: Pinch & Energy Targets, Area & Units Targets, Cost Targets.          |
| 4    | Super Targets   | Supertargeting Trade-Off Study sheet ( $\Delta T_{min}$ 0.5–40 °C, optima highlighted).                                                                            |
| 5    | Charts          | Seven live charts, one per sheet (auto-builds the trade-off study if missing).                                                                                     |

Everything except the PTA sheet is live. The Results table, the Trade-Off Study, the chart curves and the chart captions are formulas: the moment you change a stream,  $\Delta T_{min}$  or an economic parameter, they all recalculate — no button re-run needed. The PTA cascade is a snapshot (its row structure depends on the stream count); re-run Compute Targets to refresh it.

## 3. The Streams input sheet

### SnapTarget — HEN Supertargeting

Pinch targets, trade-off and PTA from the TOP v9 model | v2.6

Set  $\Delta T_{min}$  and No. of streams (the stream table resizes itself), then enter the streams; leave Type blank to auto-detect hot/cold from the temperatures. Results, the Trade-Off Study and all charts update live; re-run Compute Targets to refresh the PTA snapshot.

|                       |      |                |         |            |                          |           |
|-----------------------|------|----------------|---------|------------|--------------------------|-----------|
| $\Delta T_{min}$ (°C) | 8.0  | No. of streams |         | 6          |                          |           |
| Stream                | Type | Ts (°C)        | Tt (°C) | CP (kW/°C) | h (kW/m <sup>2</sup> °C) | Duty (kW) |
| CMIX                  | cold | 52.013         | 397     | 35.2817    | 1.066                    | 12,171.7  |
| C3                    | cold | 38             | 182     | 51.4877    | 0.58                     | 7,414.2   |
| C4                    | cold | 258            | 263     | 6.0212     | 0.58                     | 30.1      |
| H1                    | hot  | 397            | 38      | 28.2364    | 0.58                     | 10,136.9  |
| H2                    | hot  | 258            | 38      | 21.3588    | 0.58                     | 4,698.9   |
| H3                    | hot  | 164            | 38      | 25.6489    | 0.65                     | 3,231.8   |

Compute Targets

Super Targets

Charts

#### Economics

|                              |       |
|------------------------------|-------|
| Hot utility cost (\$/kW-yr)  | 128.5 |
| Cold utility cost (\$/kW-yr) | 6.426 |
| Rate of return (-)           | 0.1   |
| Plant life (yr)              | 10    |

#### Utility Temperatures

|                        |     |
|------------------------|-----|
| Hot utility Tin (°C)   | 800 |
| Hot utility Tout (°C)  | 350 |
| Cold utility Tin (°C)  | 20  |
| Cold utility Tout (°C) | 35  |

#### Area & Capital Cost

|                                                |       |
|------------------------------------------------|-------|
| Fixed cost per unit (\$)                       | 16506 |
| Area cost coefficient (\$/m <sup>2</sup> ^exp) | 145   |
| Area cost exponent (-)                         | 0.65  |

#### Utility Film Coefficients

|                                       |       |
|---------------------------------------|-------|
| Hot utility h (kW/m <sup>2</sup> °C)  | 0.111 |
| Cold utility h (kW/m <sup>2</sup> °C) | 3.75  |

The input template:  $\Delta T_{min}$  and No. of streams boxes, the self-resizing stream table (left), the four input boxes (right), action buttons.

| Input            | Location | Meaning                                                                                                                         |
|------------------|----------|---------------------------------------------------------------------------------------------------------------------------------|
| $\Delta T_{min}$ | B4       | Minimum approach temperature for targeting (°C).                                                                                |
| No. of streams   | E4       | Stream count (1–40). Changing it instantly resizes the stream table to that many rows; rows dropped when shrinking are cleared. |

| Input                     | Location | Meaning                                                                                                                                                                                                                        |
|---------------------------|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Stream rows               | A7 down  | Name, Type (hot/cold dropdown — or leave blank to auto-detect from the temperatures), Ts, Tt (°C), CP (kW/°C), film coefficient h (kW/m <sup>2</sup> ·°C). The gray Duty column shows each stream's heat load  Ts–Tt ·CP live. |
| Economics                 | J7:J10   | Hot/cold utility cost (\$/kW·yr), rate of return, plant life.                                                                                                                                                                  |
| Area & Capital Cost       | J13:J15  | Fixed cost per unit (\$), area cost coefficient (\$/m <sup>2</sup> · <sup>exp</sup> ) and area cost exponent.                                                                                                                  |
| Utility Temperatures      | J18:J21  | Hot utility Tin/Tout and cold utility Tin/Tout (°C).                                                                                                                                                                           |
| Utility Film Coefficients | J24:J25  | Hot and cold utility h (kW/m <sup>2</sup> ·°C), for heater/cooler area targets.                                                                                                                                                |

## 4. Output sheets

### 4.1 PTA — Problem Table Algorithm

#### Problem Table Algorithm

$\Delta T_{min} = 8.0 \text{ }^{\circ}\text{C}$  | shifted temperature scale (hot  $-\Delta T/2$ , cold  $+\Delta T/2$ ) | snapshot: re-run Comput

| Stream | Type | S upper ( $^{\circ}\text{C}$ ) | S lower ( $^{\circ}\text{C}$ ) |
|--------|------|--------------------------------|--------------------------------|
| CMIX   | cold | 401.0                          | 56.0                           |
| C3     | cold | 186.0                          | 42.0                           |
| C4     | cold | 267.0                          | 262.0                          |
| H1     | hot  | 393.0                          | 34.0                           |
| H2     | hot  | 254.0                          | 34.0                           |
| H3     | hot  | 160.0                          | 34.0                           |

#### Heat cascade

| S ( $^{\circ}\text{C}$ ) | $\Delta T$ ( $^{\circ}\text{C}$ ) | $\Sigma\text{CPh}-\Sigma\text{CPc}$ (kW/ $^{\circ}\text{C}$ ) | $\Delta H$ (kW) |
|--------------------------|-----------------------------------|---------------------------------------------------------------|-----------------|
| 401.0                    |                                   |                                                               |                 |
| 393.0                    | 8.0                               | -35.282                                                       | -282.3          |
| 267.0                    | 126.0                             | -7.045                                                        | -887.7          |
| 262.0                    | 5.0                               | -13.067                                                       | -65.3           |
| 254.0                    | 8.0                               | -7.045                                                        | -56.4           |
| 186.0                    | 68.0                              | 14.314                                                        | 973.3           |
| 160.0                    | 26.0                              | -37.174                                                       | -966.5          |
| 56.0                     | 104.0                             | -11.525                                                       | -1,198.5        |
| 42.0                     | 14.0                              | 23.756                                                        | 332.9           |
| 34.0                     | 8.0                               | 75.244                                                        | 602.0           |

#### PTA summary

|                          |                         |
|--------------------------|-------------------------|
| Min. hot utility QH,min  | 2,483.3 kW              |
| Min. cold utility QC,min | 934.9 kW                |
| Pinch (shifted)          | 56.0 $^{\circ}\text{C}$ |
| Hot pinch temperature    | 60.0 $^{\circ}\text{C}$ |
| Cold pinch temperature   | 52.0 $^{\circ}\text{C}$ |

Shifted stream temperatures, the full heat cascade with the pinch row highlighted, and the PTA summary.

| Column                                                        | Meaning                                                                                                                   |
|---------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| $T^*$ ( $^{\circ}\text{C}$ )                                  | Shifted interval boundary temperatures (hot streams $-\Delta T_{min}/2$ , cold streams $+\Delta T_{min}/2$ ), descending. |
| $\Delta T$ , $\Sigma\text{CPh}-\Sigma\text{CPc}$ , $\Delta H$ | Interval width, net heat-capacity flowrate of the streams present, and interval heat surplus/deficit.                     |
| Infeasible cascade                                            | Cumulative surplus with zero hot utility — the most negative value fixes QH,min.                                          |
| Feasible cascade                                              | Cascade after adding QH,min at the top; it is zero at the pinch and equals QC,min at the bottom.                          |

## 4.2 Results — full target set (live)

### HEN Targeting Results

$\Delta T_{min} = 8.0\text{ }^{\circ}\text{C}$  | SnapTarget (TOP v9 model) | live: updates with the Streams sheet

#### Pinch & Energy Targets

| Quantity                    | Value    | Units              |
|-----------------------------|----------|--------------------|
| $\Delta T_{min}$            | 8.0      | $^{\circ}\text{C}$ |
| Pinch (shifted) temperature | 56.0     | $^{\circ}\text{C}$ |
| Hot pinch temperature       | 60.0     | $^{\circ}\text{C}$ |
| Cold pinch temperature      | 52.0     | $^{\circ}\text{C}$ |
| Min. hot utility QH,min     | 2,483.3  | kW                 |
| Min. cold utility QC,min    | 934.9    | kW                 |
| Heat recovery Qrec          | 17,132.7 | kW                 |
| Total hot duty Qhot,total   | 18,067.6 | kW                 |

#### Area & Units Targets

| Quantity                      | Value   | Units        |
|-------------------------------|---------|--------------|
| Area target (counter-current) | 2,863.1 | $\text{m}^2$ |
| Area (Ft-corrected)           | 3,210.8 | $\text{m}^2$ |
| Min. units target Nu          | 10      | -            |
| Number of shells Nsh          | 21.3    | -            |
| LMTD correction Ft            | 0.892   | -            |

#### Cost Targets

| Quantity                      | Value     | Units |
|-------------------------------|-----------|-------|
| OPEX (utilities)              | \$325,118 | \$/yr |
| Ann. CAPEX (targeting)        | \$36,192  | \$/yr |
| Ann. CAPEX (design-adj.)      | \$70,202  | \$/yr |
| TAC (targeting)               | \$361,309 | \$/yr |
| TAC (design-adjusted)         | \$395,320 | \$/yr |
| Capital recovery factor (CRF) | 0.16275   | -     |

Three sections — Pinch & Energy Targets, Area & Units Targets, Cost Targets. Every quantity is a live ST\_TARGET formula on the  $\Delta T_{min}$  cell — the demo case at  $\Delta T_{min} = 8\text{ }^{\circ}\text{C}$  is shown.

### 4.3 Supertargeting Trade-Off Study (live)

#### Supertargeting Trade-Off Study (0.5-40 °C)

Targeting optimum = 0.5 °C (TAC \$295,333/yr)    Design-adjusted optimum = 8.0 °C (TAC \$395,3

*Live: the whole table, the optima and the charts recalculate when the Streams sheet changes.*

| $\Delta T_{min}$ | QH,min  | QC,min  | Qrec     | Area           | Nu | OPEX      |
|------------------|---------|---------|----------|----------------|----|-----------|
| °C               | kW      | kW      | kW       | m <sup>2</sup> | -  | \$/yr     |
| 0.50             | 1,919.0 | 370.5   | 17,697.0 | 8,898.4        | 10 | \$248,975 |
| 0.75             | 1,937.8 | 389.3   | 17,678.2 | 7,751.0        | 10 | \$251,513 |
| 1.00             | 1,956.6 | 408.1   | 17,659.4 | 7,004.0        | 10 | \$254,051 |
| 1.50             | 1,994.3 | 445.8   | 17,621.8 | 6,032.8        | 10 | \$259,127 |
| 2.00             | 2,031.9 | 483.4   | 17,584.2 | 5,395.4        | 10 | \$264,203 |
| 2.50             | 2,069.5 | 521.0   | 17,546.6 | 4,928.1        | 10 | \$269,280 |
| 3.00             | 2,107.1 | 558.6   | 17,508.9 | 4,563.2        | 10 | \$274,356 |
| 3.50             | 2,144.8 | 596.3   | 17,471.3 | 4,266.5        | 10 | \$279,432 |
| 4.00             | 2,182.4 | 633.9   | 17,433.7 | 4,018.1        | 10 | \$284,508 |
| 4.50             | 2,220.0 | 671.5   | 17,396.1 | 3,805.8        | 10 | \$289,584 |
| 5.00             | 2,257.6 | 709.1   | 17,358.4 | 3,621.3        | 10 | \$294,661 |
| 6.00             | 2,332.9 | 784.4   | 17,283.2 | 3,314.4        | 10 | \$304,813 |
| 7.00             | 2,408.1 | 859.6   | 17,208.0 | 3,067.4        | 10 | \$314,965 |
| 8.00             | 2,483.3 | 934.9   | 17,132.7 | 2,863.1        | 10 | \$325,118 |
| 9.00             | 2,558.6 | 1,010.1 | 17,057.5 | 2,690.6        | 10 | \$335,270 |
| 10.00            | 2,633.8 | 1,085.3 | 16,982.2 | 2,542.5        | 10 | \$345,422 |
| 11.00            | 2,709.1 | 1,160.6 | 16,907.0 | 2,413.6        | 10 | \$355,575 |
| 12.00            | 2,784.3 | 1,235.8 | 16,831.7 | 2,300.4        | 10 | \$365,727 |
| 13.00            | 2,859.6 | 1,311.1 | 16,756.5 | 2,200.0        | 10 | \$375,880 |
| 14.00            | 2,934.8 | 1,386.3 | 16,681.2 | 2,110.3        | 10 | \$386,032 |
| 15.00            | 3,010.1 | 1,461.6 | 16,606.0 | 2,029.7        | 10 | \$396,184 |
| 16.00            | 3,085.3 | 1,536.8 | 16,530.8 | 1,956.7        | 10 | \$406,337 |
| 18.00            | 3,235.8 | 1,687.3 | 16,380.3 | 1,829.8        | 10 | \$426,642 |
| 20.00            | 3,386.3 | 1,837.8 | 16,229.8 | 1,723.2        | 10 | \$446,946 |
| 22.00            | 3,536.8 | 1,988.3 | 16,079.3 | 1,632.5        | 10 | \$467,251 |
| 24.00            | 3,687.3 | 2,138.8 | 15,928.8 | 1,554.4        | 10 | \$487,556 |
| 26.00            | 3,837.7 | 2,289.2 | 15,778.3 | 1,486.4        | 10 | \$507,861 |
| 28.00            | 3,988.2 | 2,439.7 | 15,627.8 | 1,426.8        | 10 | \$528,165 |
| 30.00            | 4,138.7 | 2,590.2 | 15,477.3 | 1,374.2        | 10 | \$548,470 |
| 32.00            | 4,289.2 | 2,740.7 | 15,326.9 | 1,327.5        | 10 | \$568,775 |
| 34.00            | 4,439.7 | 2,891.2 | 15,176.4 | 1,285.7        | 10 | \$589,080 |
| 36.00            | 4,590.2 | 3,041.7 | 15,025.9 | 1,248.2        | 10 | \$609,384 |
| 38.00            | 4,740.7 | 3,192.2 | 14,875.4 | 1,214.5        | 10 | \$629,689 |
| 40.00            | 4,891.2 | 3,342.7 | 14,724.9 | 1,184.4        | 10 | \$649,994 |

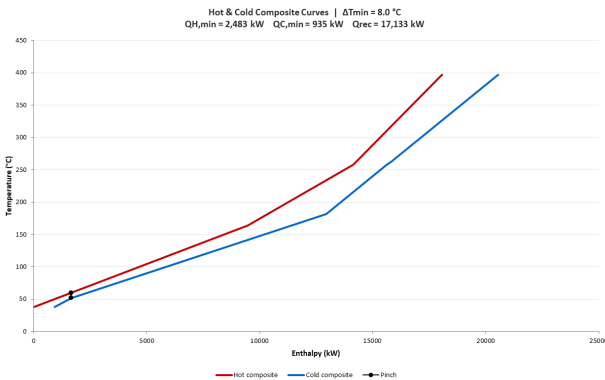
$\Delta T_{min}$  sweep 0.5–40 °C. Green row = targeting-level optimum (min TAC with pure counter-current area); blue row = design-adjusted optimum (min TAC with  $F_t$  + shells correction). The highlighting is conditional formatting, so it moves automatically when the inputs change.

| Column group           | Meaning                                                                            |
|------------------------|------------------------------------------------------------------------------------|
| QH,min / QC,min / Qrec | PTA energy targets and process-to-process heat recovery at each $\Delta T_{min}$ . |

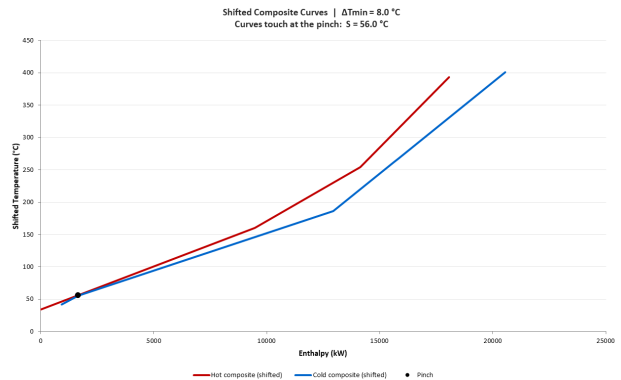
| Column group                | Meaning                                                                                  |
|-----------------------------|------------------------------------------------------------------------------------------|
| Area / Nu                   | Bath-formula counter-current area target and minimum number of units (Euler, MER).       |
| OPEX / ACAPEX (t) / TAC (t) | Utility cost, annualized capital and total annualized cost of the targeting-level model. |
| Nsh / Ft / Area (Ft)        | Estimated 1–2 shells, average LMTD correction factor, Ft-corrected area.                 |
| ACAPEX (d) / TAC (d)        | Annualized capital and TAC of the design-adjusted model (shells + Ft).                   |

## 4.4 Charts (live, one per sheet)

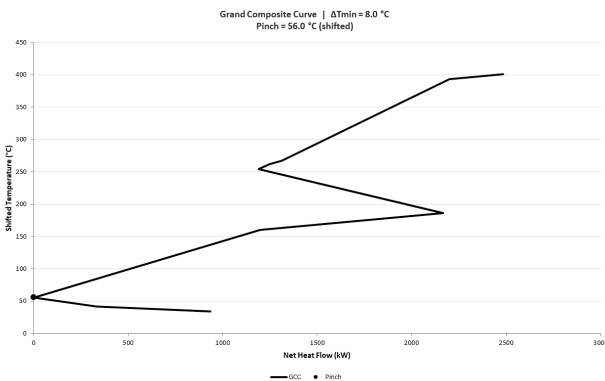
Each chart lives on its own sheet — Composite Curves, Shifted Composites, Grand Composite, Trade-Off (Targeting), Trade-Off (Design), Energy Targets, Area Targets — so the workbook stays organized. All seven read their data and captions from formulas: titles show the current QH/QC/Qrec, pinch and optima and follow every input change.



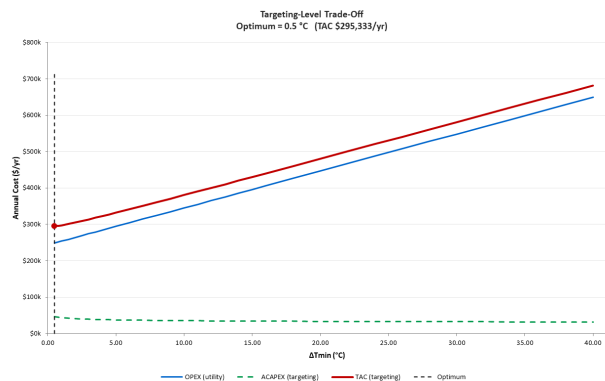
Composite curves with the pinch gap marked



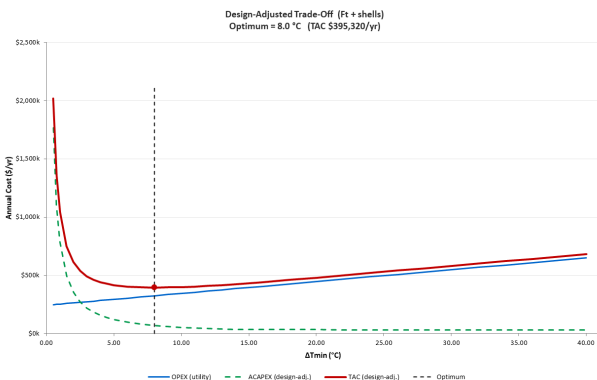
Shifted composite curves touching at the pinch



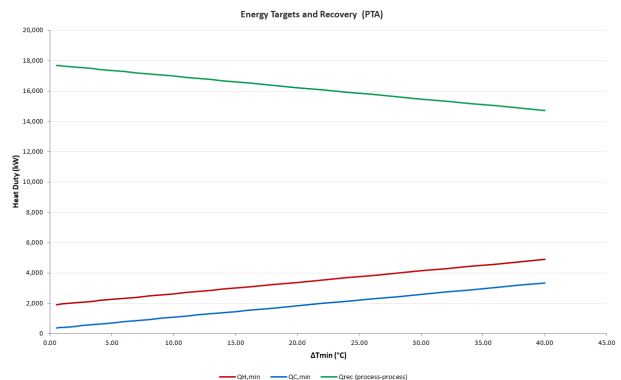
Grand composite curve with the pinch point



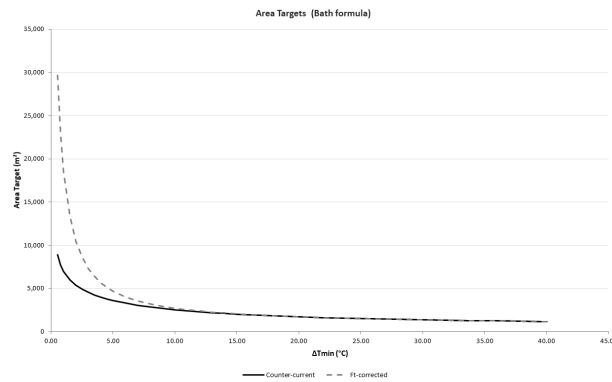
Targeting-level trade-off with its optimum



Design-adjusted trade-off with its optimum



Energy targets vs  $\Delta T_{min}$



Area targets vs  $\Delta T_{min}$

## 5. Worksheet function reference

All functions read the stream table and economics from the Streams sheet of the workbook they are used in, and recalculate automatically (volatile). They work in any workbook created by New Template.

| Function               | Returns                                                                                                                                                   | Example                          |
|------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------|
| ST_TARGET(dtmin, key)  | Any single target at the given $\Delta T_{min}$ — see the key table below.                                                                                | =ST_TARGET(8, "tac_d") → 395,320 |
| ST_QHMIN(dtmin)        | Minimum hot utility (kW).                                                                                                                                 | =ST_QHMIN(8) → 2,483.3           |
| ST_QCMIN(dtmin)        | Minimum cold utility (kW).                                                                                                                                | =ST_QCMIN(8) → 934.9             |
| ST_PINCH(dtmin)        | Pinch temperature, shifted scale (°C).                                                                                                                    | =ST_PINCH(8) → 56.0              |
| ST_AREA(dtmin)         | Counter-current area target (m²).                                                                                                                         | =ST_AREA(8) → 2,863.1            |
| ST_UNITS(dtmin)        | Minimum number of units (—).                                                                                                                              | =ST_UNITS(8) → 10                |
| ST_TAC(dtmin, [model]) | Total annualized cost; model "d" (default, design-adjusted) or "t" (targeting).                                                                           | =ST_TAC(8, "t") → 361,309        |
| ST_OPT(model, what)    | Optimum of the 0.5–40 °C sweep; model "t"/"d"; what "dt", "tac" or "tacmax".                                                                              | =ST_OPT("d", "dt") → 8.0         |
| ST_PINCHH(dtmin)       | Hot-composite enthalpy at the hot pinch temperature (kW) — x-position of the pinch marker.                                                                | =ST_PINCHH(8)                    |
| ST_CURVE(kind, dtmin)  | 60×2 curve block (X, Y) padded with #N/A; kind "hot", "coldshift", "gcc", "hot_s" or "cold_s" (shifted composites). Enter over a 60-row × 2-column range. | =ST_CURVE("gcc", 8)              |

### Keys accepted by ST\_TARGET

| Key        | Quantity                        | Units | Key        | Quantity                  | Units |
|------------|---------------------------------|-------|------------|---------------------------|-------|
| qh         | Min. hot utility QH,min         | kW    | opex       | Utility operating cost    | \$/yr |
| qc         | Min. cold utility QC,min        | kW    | acapex_t   | Ann. capital, targeting   | \$/yr |
| qrec       | Heat recovery (process–process) | kW    | acapex_d   | Ann. capital, design-adj. | \$/yr |
| qhot_total | Total hot-stream duty           | kW    | tac_t      | TAC, targeting            | \$/yr |
| area       | Area target (counter-current)   | m²    | tac_d      | TAC, design-adjusted      | \$/yr |
| area_ft    | Area, Ft-corrected              | m²    | pinch      | Pinch (shifted)           | °C    |
| nu         | Min. units (Euler MER)          | —     | pinch_hot  | Hot pinch temperature     | °C    |
| nsh        | Number of 1–2 shells            | —     | pinch_cold | Cold pinch temperature    | °C    |
| ft         | LMTD correction factor          | —     | crf        | Capital recovery factor   | —     |

## 6. Reading the results — demo case

The two cost models bracket reality: the targeting-level model assumes pure counter-current exchangers at the minimum unit count, so its optimum sits at a very small  $\Delta T_{\min}$ ; the design-adjusted model adds the  $F_t$  correction and a shells estimate, which penalizes small approach temperatures and moves the optimum to a practical value — in agreement with the Aspen Energy Analyzer result ( $\approx 9^\circ\text{C}$ ) for this plant.

| Quantity (kerosene hydrotreater demo)  | Value                                                                                                         |
|----------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Targeting-level optimum                | $\Delta T_{\min} = 0.5^\circ\text{C}$ , TAC = \$295,333/yr                                                    |
| Design-adjusted optimum                | $\Delta T_{\min} = 8.0^\circ\text{C}$ , TAC = \$395,320/yr                                                    |
| At $\Delta T_{\min} = 8^\circ\text{C}$ | QH,min = 2,483.3 kW, QC,min = 934.9 kW, Qrec = 17,132.7 kW                                                    |
| Pinch at $8^\circ\text{C}$             | 56.0 $^\circ\text{C}$ shifted (hot 60.0 $^\circ\text{C}$ / cold 52.0 $^\circ\text{C}$ )                       |
| Area / units at $8^\circ\text{C}$      | 2,863.1 m <sup>2</sup> counter-current (3,210.8 m <sup>2</sup> Ft-corrected), Nu = 10, Nsh = 21.3, Ft = 0.892 |
| Costs at $8^\circ\text{C}$             | OPEX \$325,118/yr, ACAPEX \$36,192 (t) / \$70,202 (d), CRF = 0.16275                                          |

## 7. Model and validation

Energy targets: Problem Table Algorithm on shifted temperatures. Area target: Bath formula (vertical heat-transfer model) on balanced composite curves with the Chen LMTD approximation and stream-individual film coefficients. Units target: Euler ( $N-1$ ) with MER pinch decomposition. Design adjustment: shells power-law (reference  $15^\circ\text{C}$ , exponent 1.2) and  $F_t = 1 - 0.8 \cdot \exp(-0.25 \cdot \Delta T_{\min})$  floored at 0.30 (Ahmad–Linnhoff–Smith supertargeting). Economics: CAPEX =  $N \cdot (16,506 + 145 \cdot A_{\text{per}}^{0.65})$  \$, annualized with CRF at 10 % over 10 years; utilities 128.5 / 6.426 \$/kW·yr.

The add-in is a line-by-line port of the TOP v9 Python model (trade\_off\_plot\_9.py), whose energy and area targets are validated against OpenPinch (energy within 0.04 %, area within 6 %). The VBA engine reproduces the full Python sweep — 34  $\Delta T_{\min}$  points  $\times$  14 quantities — to a worst relative deviation of  $4 \times 10^{-15}$  (machine precision).

## 8. Licence

SnapTarget is proprietary software, licensed — not sold. Subject to payment of the applicable licence fee, you are granted a personal, non-exclusive, non-transferable licence to install and use it for your own internal purposes, granted per named user: you may install it on any number of computers you alone control, but the licence may not be shared, pooled or used concurrently by more than one person. Until a licence fee has been paid you may use it only to evaluate it — privately, non-commercially, and not for any result that informs a real design, purchase, tender, report or deliverable.

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